MIL-M-80226A
26 November 1984
SUPERSEDING
MIL-M-80226
16 August 1971

MILITARY SPECIFICATION

METALLIZING SYSTEM, WIRE, ELECTRIC ARC, GUNS AND ACCESSORIES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers metallizing systems which apply metallic coatings in the form of molten spray. The molten metal spray is developed by an electric arc generated between the ends of two electrically charged wires. High velocity compressed air or other suitable gases atomizes the molten metal and projects it onto the workpiece to be coated.

1.2 Classification. This specification covers metallizing systems of the following sizes: The size to be furnished shall be as specified (see 6.2.1).

Size 1 - 200 amperes at 20 to 30 volts
Size 2 - 350 amperes at 20 to 35 volts
Size 3 - 500 amperes at 20 to 40 volts

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or letter.

FSC 3433

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SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies, and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/IEEE 268 - Metric Practice.

ANSI/NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Commercial Packaging, Standard Practice for.

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(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Design. The metallizing system shall be new and one of the manufacturer's current models capable of spraying molten metals developed in an arc by simultaneously melting and atomizing two electrically charged wires. The system shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. The metallizing system shall be designed and constructed so that it can be shipped fully assembled and ready for use when connected to the proper electric and pneumatic services.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the system. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in U.S. units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in ANSI/IEEE 268.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the system shall be graduated in the same system.

3.2.2 Reclaimed materials. The system may contain reclaimed materials provided such materials will not jeopardize the system's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the system.

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3.2.3 Energy efficiency. The metallizing system and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the Industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. The metallizing system shall be equipped with safety devices as required to protect personnel from electrical voltages constituting a safety hazard. Ground terminals shall be provided on the chassis of the control console and power supply. Power supply terminals shall be enclosed in a manner that will prevent accidental contact by operating personnel. The entire outer shell of the spray gun assembly shall be electrically neutral or insulated to insure that the operator will not be exposed to electrically charged components. Equipment safety devices shall include appropriate devices to protect the direct current power supply from overload. All system parts, components, mechanisms, and assemblies furnished on the system, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the system itself. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.7 Interchangeability. To provide for replacement of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or system.

3.3 Construction. The metallizing system shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the system. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

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3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the metallizing system shall be painted in accordance with the manufacturer's commercial practice, provided that all surfaces to be painted shall be cleaned of all foreign matter and not less than one coat of primer and one coat of finish color is applied.

3.3.6 Threads. All threaded parts used on the system and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with the requirements of MIL-STD-461.

3.4 Components. The metallizing system shall consist of at least the following components.

3.4.1 Gun. The spray gun shall be designed to melt and spray 11 or 14 gauge (B&S) wire (.091 or .064 inch diameter). Wire guides, that also serve as electrodes, shall be provided to accurately guide the wires to the converging point where the arc is established. Electric power to the wires shall be supplied through the electrodes. The design of the electrodes shall provide for minimum wear and heat abrasion, and for maximum transfer of electric power to the wires. The compressed air atomizing nozzle shall be positioned for maximum concentration and efficiency of the spray pattern. The gun shall be equipped with all fittings and connections required for feeding the wire into the gun and through the wire guides. All necessary connections shall be provided for connection of electric power, compressed air or gas, and control circuits. A gun mounted control or controls shall start and stop air or gas, wire feed, and arc current. The gun shall be designed for machine mounting or hand use. The maximum weight of the gun, not containing wire drive units, shall not exceed 5 pounds. The weight of the gun, containing built-in wire drive units, shall not exceed 20 pounds. The above weights are exclusive of electrical cables, wire feed conduits, and air hose. The rated continuous duty DC power capability of the gun

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shall be not less than that required for operation with the full DC power output capability of the power supply.

3.4.2 Wire feed mechanism. The system shall include a dual wire feed mechanism which may be an integral part of the spray gun or it may be a separate unit. The wire feed mechanism shall be powered by an infinitely variable speed electric motor, air motor, or air turbine. The motor or turbine shall have adequate power to feed the wire at a constant rate from fully loaded reels at any speed within the range of the feed unit. Wire feed rates shall be provided to meet all spraying rates listed in table I. The drive roll pressure shall be adjustable to assure positive wire feed. The drive roll mechanism shall be insulated from the gun body or case.

TABLE I. Spray capacity at continuous duty.

Size	Pounds per hour (minimum)			
	Aluminum	Stainless	Zinc	Steel
1	11	16	22	18
2	21	35	84	35
3	25	48	115	50

3.4.2.1 Wire reels and stand. The wire feed mechanism shall have two separate reels for holding wire coils. Each reel shall be mounted on a stand and readily accessible for easy loading. The reels shall be electrically insulated from the stand and from each other. A friction brake on each reel shall control the tension of the wire drag. The wire straighteners for each reel shall be rigidly fastened to and insulated from the reel stand. The straighteners shall be capable of removing the set curvature from any coiled or spooled wire used with the system. The stand shall be mounted on casters and shall be partially enclosed. Each reel shall be capable of holding and operating with a wire coil weight of not less than 50 pounds.

3.4.2.2 Wire feed controls. The wire feed controls shall be located in a convenient and accessible position on the gun or control panel. The wire feed rate shall be adjustable manually to any speed within the operating range of the feed system. The controls shall provide fine adjustment of the rate of wire feed and shall include an instant "stop-start" mechanism that does not change the feed rate. The system shall prevent fusing of the wires when spraying is stopped.

3.4.3 Power supply (DC). The power supply shall be of the transformer-rectifier, or the motor-generator DC type. The power supply shall be designed for operation with electric arc wire metallizing equipment and shall be regulated as necessary to provide DC power having the required constant-voltage characteristic. The rated continuous duty DC output current (100% duty cycle) and voltage shall be as specified in table II. The output voltage shall be continuously variable throughout the range of the power supply. The power supply shall be either wheel or skid mounted.

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TABLE II. Power supply output.

Size	Amperes 1/	Voltage range 2/
1	200	20 to 30
2	350	20 to 35
3	500	20 to 40

1/ The amperes shall be not less than the value stated.
2/ Where a range is shown, the voltage shall be not greater than the lowest figure to not less than the highest figure.

3.4.3.1 Transformer-rectifier type power supply. The transformer-rectifier power supply shall provide a DC constant voltage output as described in paragraph 3.4.3. The power supply shall provide three phase full wave rectification which shall be accomplished with solid state rectifiers. A suitably rated contactor shall be furnished for use in the primary input power leads. The power supply shall be enclosed in a cabinet of steel frame and panel contruction. The gage of the metal, and the physical strength of the chassis and cabinet shall be adequate for support and protection of all components. The side and top panels of the cabinet shall be removable to provide access for servicing. Openings shall be provided for adequate ventilation. Forced air cooling shall be provided by a fan mounted inside the cabinet. The fan shall be positioned so that the air flow will be sufficient for maintaining a safe operating temperature, under full rated load conditions in an ambient temperature of 100 degrees Fahrenheit (F). The fan motor shall be designed to operate from the same electric power source as the power supply.

3.4.3.2 Motor-generator (MG) power supply. The motor-generator power supply shall consist of an alternating current operated motor driving a DC generator. The MG set shall be a horizontal type and shall be totally enclosed except for openings for cooling air intake and exhaust. The motor and generator rotors shall be mounted on the same shaft. The motor shall operate from a three phase, 60 Hz, alternating current power source, and the generator output shall be DC with a voltage and current output rating as specified (see 3.4.3). The bearings shall be grease or oil type, ball or roller, construction, and selected in accordance with the bearing manufacturer's recommendations covering application, speed, and loading. The design of the MG set shall include ease of maintenance of all components as well as features to minimize damage to rotor or the stator due to bearing failure. The motor starter shall be of the full voltage, across the line magnetic type with overload protection, reset button,, and a start and stop push button control. The starters shall have a rating suitable for use with the MG set. Forced air cooling shall be accomplished by a fan mounted on the armature shaft. Cooling shall be adequate for continuous operation under full rated load in an ambient temperature of 100 degrees F.

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3.4.3.3 Power supply controls. The power supply controls shall include an "on-off" or starting switch and an output voltage control. A DC voltmeter and ammeter for measuring the output voltage and current shall be included and mounted on the control panel or power supply unit.

3.4.4 Air filtration and control. The metallizing system shall be equipped with an air filtration and control system. The air filtration system shall include a cleanable or replaceable air filter and an oil-water separator. The air control system shall be equipped with a control valve, pressure regulator, air pressure gauges for atomizing air, and when applicable, controls for the wire feed air motor or air turbine. An automatic means shall be provided to prevent application of electric power to the spray gun, when air pressure is less than that required for safe operation of the system as specified by the manufacturer. The air filtration and control equipment may be located either in a separate unit or be an integral part of the wire control unit or power supply.

3.4.5 Hose and power cables. All necessary hoses and power cables for interconnecting the metallizing system shall be furnished. The hoses and power cables shall be not less than 15 feet in length. Insulated flexible conduit shall be furnished to encase the wires between reel stands or straighteners and the spray unit. The flexible conduit shall be not less than 7-1/2 feet in length.

3.4.6 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 70. The metallizing system shall operate from a single 230/460, 3 phase, 60 Hz circuit. The system shall be initially wired for 230 volt operation.

3.4.6.1 Motors. Motors shall be rated for continuous duty operation, with horsepower and speed rating as required for maximum load conditions.

3.4.6.2 Solid state components. Solid state design shall be used throughout for electronic components.

3.4.6.3 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 115 volts.

3.4.7 Air or gas supply. The metallizing system shall operate at maximum rated capacity from an air or gas source which shall not exceed 60 cubic feet per minute (CFM) and a maximum pressure of 90 pounds per square inch (PSI).

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3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the system shall be not less than those specified in table I and II for the system size specified.

3.6 Performance. The metallizing system shall be capable of melting and spraying in wire form steel, stainless steel, zinc, and aluminum. The deposition rates shall be not less than that specified in table I.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. Unless otherwise specified one set of hand tools, normally furnished by the manufacturer, shall be furnished. Additional equipment required shall be furnished as specified and fully described (see 6.2.1).

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.10 Marking of instruction plates, indicating plates, and name plates. All words on instruction plates, indicating plates, and name plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. When these methods are not practicable, the instructions shall be applied directly on the item by environmentally protected decalcomania transfer, metal wrap-around tag, stencil, silk screen, or by any other method suitable for the use intended which shall be as permanent as the normal life expectancy of the item to which it is applied and capable of withstanding the environmental tests and cleaning procedures specified for the item.

3.10.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be furnished. If a plate is furnished it shall be permanently and securely attached to each metallizing system. If a chart is furnished it may be part of the instruction manual or it may be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each metallizing system. The nameplate shall contain the information listed below. If the metallizing system is a special model, the model designation shall include the model of the basic

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standard system and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amperes, phase, frequency)
Rated DC output - volts
Rated DC output - current
Compressed air requirements
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Hour meter. When specified (see 6.2.1), the metallizing system shall be equipped with an hour meter. The hour meter shall operate whenever the system's main power switch is energized and shall display accumulated operating time. The meter shall be of the non-resetting type and shall have a range of 0 to 99,999 hours. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the system. Upon reaching the maximum accumulative hours, the meter readout shall automatically revert to zero and continue to totalize time.

3.12 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be in the English language.

3.13 Workmanship. Workmanship of the metallizing system and accessories shall be of a quality equal to that of the manufacturer's standard equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the item to pass the first article examination and tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1) quality conformance inspection shall consist of the examination in 4.5, the test in 4.6.1, and the inspection in 4.7. Failure of the item to pass the examination, test and inspection shall be cause for rejection.

4.5 Examination. The system shall be visually examined to determine compliance with the requirements of 3.2 through 3.5 and 3.7 through 3.13.

4.6 Tests.

4.6.1 Operational test. The metallizing system shall be connected to the specified electric and air sources and shall be tested under operating conditions for not less than 10 minutes. The wire used during this test shall be as determined by the equipment manufacturer. Performance of the spray gun, wire drive mechanism, power supply, compressed air system, and all controls and instrumentation shall be checked for proper operation. Any evidence of faulty or erratic operation shall be cause for rejection.

4.6.2 Performance test. The metallizing system shall be tested for metallizing capabilities. The test shall consist of using wire made of each of the materials specified in table I. Four specimens shall be sprayed, one specimen for each type of wire specified. Preparation of the specimens and the spray technique shall be the responsibility of the manufacturer. The resulting coating on each specimen shall be visually examined. The coating shall be uniform in thickness, with no lumps or spatter, or other evidence of system malfunction. Sufficient spraying shall be accomplished to verify that the spray rate for each material is not less than that specified in table I.

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4.6.3 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.3.7.

4.7 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Packaging, packing, and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or B preservation, level A or B packing, and special marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The metallizing systems described herein are intended for use in facilities having a requirement for medium or high speed metal spraying capability.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. When first article is required for inspection and approval (see 3.1).

d. If system is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Painting, if different (see 3.3.5).

h. Electromagnetic interference control, if required (see 3.3.7).

i. Electrical system and input power supply, if different (see 3.4.6).

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j. Size and capacity, if different (see 3.5).

k. Additional performance features, if required (see 3.6.1).

l. Standard equipment, if different (see 3.7).

m. Specify and fully describe optional equipment required, if different (see 3.8).

n. Specify applicable repair parts, if required (see 3.9).

o. Lubrication chart or plate, if different (see 3.10.1).

p. Nameplate, if different (see 3.10.2).

q. Hour meter, if required (see 3.11).

r. First article inspection, if different (see 4.3).

s. Quality conformance inspection, if different (see 4.4).

t. If military packaging, packing and marking is required, specify the level required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operators manuals, parts lists, wiring diagrams, and examination, original alignment, and tolerance test reports and other instructions for operation and maintenance should be specified on a DD Form 1423, Contract Data Requirements List incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the system itself. It does not require compliance with those OSHA requirements that concern "the system in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual systems in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional system requirements that will integrate the new system into its future operating environment. The system shall be equipped with all points-of operation guarding normally furnished as standard on the manufacturer's commercial

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systems supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1(f).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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